psgscoring

A Technical Handbook for Automated Polysomnography Scoring

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Abstract

This handbook provides a complete technical introduction to psgscoring, an open-source Python library (BSD-3) for AASM 2.6-compliant automated polysomnography (PSG) scoring. It covers clinical context, signal processing, event detection, apnea classification, twelve bias corrections, hypoxic burden computation, post-processing for central apnea refinement, and external validation.

Prerequisites: Python 3.9+, NumPy, SciPy, basic DSP (Fourier transform, filtering, envelopes).
Repository: https://github.com/bartromb/psgscoring
Installation: pip install psgscoring

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1. Clinical Context: Why This Matters

1.1 Sleep Apnea

Obstructive Sleep Apnea (OSA) affects approximately 936 million adults worldwide. During sleep, the upper airway collapses partially or completely, causing breathing interruptions that fragment sleep and reduce blood oxygen levels. Left untreated, OSA increases the risk of hypertension, stroke, heart failure, and traffic accidents.

Diagnosis relies on polysomnography (PSG): an overnight recording of brain waves (EEG), eye movements (EOG), muscle tone (EMG), airflow, respiratory effort, blood oxygen saturation (SpO_2), body position, snoring, heart rhythm (ECG), and leg movements.

1.2 The Scoring Problem

A sleep technologist manually reviews the recording and counts:

- **Apneas**: $\geq$90% flow reduction for $\geq$10 s
- **Hypopneas**: $\geq$30% flow reduction for $\geq$10 s with $\geq$3% oxygen desaturation or an EEG arousal

The Apnea-Hypopnea Index AHI = (apneas + hypopneas)/hours of sleep is the primary diagnostic metric. Manual scoring takes 1–2 hours per study with 10–20% inter-scorer variability. `psgscoring` automates this process.

1.3 The AASM Standard

The American Academy of Sleep Medicine (AASM) Manual for the Scoring of Sleep and Associated Events (Version 2.6, 2020) is the rulebook [1]. Key rules: Rule 1A: hypopnea requires $\geq$3% SpO_2 desaturation. Rule 1B: hypopnea may alternatively be scored with an EEG arousal. Apneas are scored on the thermistor; hypopneas on the nasal pressure transducer.

1.4 Beyond AHI: Why One Number Is Not Enough

The AHI, though ubiquitous, has been critically appraised as a reductionist metric. Pevernagie et al. demonstrated that the correlation between AHI and clinical endpoints—sleepiness, hypertension, quality of life—is surprisingly weak, and that reliance on the AHI leads to overestimation of OSA's clinical relevance [3]. Two patients with identical AHI values can experience fundamentally different physiological consequences [4].

`psgscoring` addresses this by computing three complementary dimensions: (1) AHI confidence interval (event frequency with uncertainty), (2) hypoxic burden (oxygen deficit per hour), and (3) arousal index (sleep fragmentation).

2. Architecture Overview

`psgscoring` is organised as 13 independently importable submodules with a strict one-directional dependency graph:

Table 1: Submodule responsibilities (v0.2.93)

Module	Responsibility	LOC
`constants`	AASM thresholds, band limits, scoring profiles	76
`utils`	Sleep mask construction, channel detection	134
`signal`	Linearisation, baseline, MMSD validation	327
`breath`	Breath segmentation, flattening index	254
`classify`	7-rule apnea type decision tree	310
`spo2`	SpO_2 coupling, ODI, hypoxic burden	395
`ecg_effort`	ECG-derived effort (TECG + spectral classifier)	180
`plm`	Periodic Limb Movement detection (AASM + WASM)	271
`ancillary`	Heart rate, snoring, position, Cheyne-Stokes	277
`signal_quality`	Per-channel quality grading	160
`postprocess`	CSR reclassification, mixed decomposition, CII	310
`respiratory`	Apnea/hypopnea detection + 12 corrections	1061
`pipeline`	Master orchestrator (11 steps)	802

The public API exports 42 symbols from `psgscoring/__init__.py`.

3. Signal Processing Chain

3.1 Step 1: Nasal Pressure Linearisation

The nasal pressure transducer measures pressure, not flow. The Bernoulli relation gives $x_{\text{lin}}(t) = \text{sign}(x(t)) \cdot \sqrt{|x(t)|}$.

```
def linearise_nasal_pressure(signal):
    return np.sign(signal) * np.sqrt(np.abs(signal))
```

3.2 Step 2: Bandpass Filtering

A 4th-order Butterworth bandpass (0.05–3 Hz) removes DC drift and high-frequency noise. The normal respiratory rate (12–20/min = 0.2–0.33 Hz) falls well within this passband.

3.3 Step 3: Hilbert Envelope

The instantaneous amplitude: $A(t) = |x(t) + i\mathcal{H}\{x(t)\}|$. This provides a smooth, non-negative representation of breath amplitude. The envelope smooths over individual breath cycles—useful for apneas but can miss subtle hypopneas (motivating peak-based detection in Section 7).

3.4 Step 4: Dynamic Baseline

A 5-minute sliding window, 95th percentile, advancing in 10-second steps with linear interpolation: $B(t) = P_{95}\{A(\tau) : \tau \in [t - 150\text{s}, t + 150\text{s}]\}$.

3.5 Step 5: Event Detection

An event is detected when normalised amplitude drops below threshold for ≥10 s: $A(t)/B(t) \leq 1 - \theta$ where $\theta = 0.90$ for apneas, $\theta = 0.30$ for hypopneas.

3.6 Step 6: MMSD Validation

The Mean Magnitude of Second Derivative $\overline{|x''(t)|}$ distinguishes true apneas (low but non-zero MMSD from cardiac oscillation) from artefacts (MMSD ≈ 0).

4. Apnea Type Classification

A 7-rule decision tree classifies each event as obstructive, central, or mixed based on thoraco-abdominal effort signals.

4.1 Feature Extraction

For each event: `effort_ratio` (RMS during event / baseline), `raw_var_ratio`, `paradox_corr` (Pearson of thorax vs. abdomen), `phase_angle_deg` (Hilbert phase difference), `quarter_efforts`, `flattening_index`.

4.2 Decision Tree

Table 2: Classification rules (first match wins)

Rule	Condition	Type	Conf.
0	Phase angle $\geq 45°$	Obstructive	0.75–0.97
1	Paradoxical + high variability	Obstructive	0.85–0.95
2	High variability, low envelope	Obstructive	0.70–0.85
3	First half low, second half high	Mixed	0.70–0.95
4	Clear effort present	Obstructive	0.65–0.80
5/5a/5b	No effort (+ ECG confirmation)	Central	0.45–0.90
6	Default (low effort $\rightarrow$ central; else obstr.)	varies	<0.60

The cardiac pulsation problem: during a true central apnea, the heart continues beating. These mechanical oscillations appear on the RIP bands as low-amplitude signals. Rules 5/5a/5b use relaxed thresholds and ECG-derived effort to handle this.

5. SpO_2 Coupling

AASM Rule 1A requires $\geq$3% desaturation. The algorithm computes the SpO_2 baseline (90th percentile during sleep), searches for the nadir in a post-event window (30–45 s, configurable), and checks desaturation $\geq$3%.

Circulatory delay: finger oximetry has 20–40 s delay. The standard profile uses 45 s; strict uses 30 s.

6. Over-Counting Corrections

Six mechanisms that inflate AHI:

Fix 1: Post-Apnea Baseline Inflation. Recovery hyperpnea enters the baseline window, inflating it. Fix: exclude a 30-second recovery mask after each apnea.

Fix 2: SpO_2 Cross-Contamination. At AHI >60/h, the nadir of event N overlaps event $N+1$'s window. Fix: flag overlapping events (informational since v0.8.14).

Fix 3: Cheyne-Stokes Flagging. Each CSR trough crosses the 30% threshold. Fix: detect periodicity via autocorrelation; flag matching events as `csr_flagged`.

Fix 4: Confidence Stratification. Poor RIP quality defaults to obstructive. Fix: report events at four confidence tiers ($\geq$0.85, $\geq$0.60, $\geq$0.40, all).

Fix 5: Artefact-Flank Exclusion. Signal recovery ramp after a flat-line gap. Fix: append 15-second exclusion window after each gap ≥10 s.

Fix 6: Local Baseline Validation. Rolling baseline inflated by recovery hyperpnea. Fix: compare event amplitude with 30-second pre-event window; reject if local reduction <20% (raised to 30% when surrounding breathing is stable, CV <0.30).

7. Under-Counting Corrections

Six mechanisms that suppress detection:

Fix 7: Peak-Based Breath Detection. AASM requires *peak* excursion ≥30%. The Hilbert envelope smooths over individual breaths. Fix: detect breaths via zero-crossings, compute per-breath amplitude, combine with envelope mask via logical OR. Requires ≥3 consecutive reduced breaths (anti-fragmentation).

Fix 8: Extended SpO_2 Window. 30 s misses late nadirs. Fix: extend to 45 s (configurable).

Fix 9: Flow Smoothing. *Removed.* Ablation analysis on PSG-IPA showed that 3-second smoothing was the dominant source of over-counting (+54 false hypopneas on a single mild recording). The consecutive-breath requirement (Fix 7) now replaces smoothing as the anti-fragmentation mechanism.

Fix 10: Position Signal Auto-Mapping. Some EDF systems store position as raw ADC values (0–255). Fix: auto-detect encoding; apply percentile-based quantisation.

Fix 11: Configurable Scoring Profiles. Three profiles balance sensitivity vs. specificity:

Table 3: Scoring profiles

Profile	Thresh.	SpO_2 win.	Peak det.	Stab. filter	Use case
Strict	≥30%	30 s	Envelope only	CV<0.45	Research
Standard	≥30%	45 s	Peak+envelope	CV<0.45	Clinical
Sensitive	≥25%	45 s	Peak+envelope	CV<0.45	UARS

Fix 12: Flattening-Based RERA Detection. The flattening index FI measures the fraction of inspiratory time exceeding 80% of peak amplitude. Normal ≈0.10; flow-limited >0.30. Sequences of ≥3 breaths with FI >0.30 spanning ≥10 s, terminated by an arousal, are scored as flattening-RERAs. RDI = AHI + (FRI-RERA + flat-RERA)/TST.

8. Breath-Amplitude Stability Filter

At low event density, normal breath-to-breath variability generates apparent ≥30% reductions. For each hypopnea candidate, the system examines breath amplitudes in a ±120-second window. If the coefficient of variation CV <0.45 with ≥10 breaths available, the candidate is rejected as normal variability.

Calibrated on PSG-IPA: rejected 56 false-positive hypopneas in a normal recording (SN4: −68%) while preserving accuracy in severe OSA (SN3: −3%).

9. AHI Confidence Interval

Every analysis runs all three profiles and reports the AHI as an interval with a robustness grade: **A** (all profiles same severity), **B** (2/3 concordant), **C** (all discordant).

```
results = run_pneumo_analysis(raw, hypno, channel_map)
iv = results["ahi_interval"]
print(f"AHI: {iv['standard']['ahi']}")
```

```
print(f"Interval: [{iv['strict']['ahi']}"
      f"-{iv['sensitive']['ahi']}]")
print(f"Grade: {iv['robustness_grade']}")
```

Clinical example (SN2): 12 technologists split 8/4 between Normal and Mild. Algorithm: interval [4.9–8.9], Grade B. The interval immediately communicates that the AHI hovers near the 5/h threshold.

10. ECG-Derived Effort Classification

Spectral classifier: PSD of the RIP signal during each event. If >75% of power in the cardiac band (0.8–2.5 Hz) and <20% in respiratory (0.1–0.5 Hz) → flag as probable central.

TECG (Berry et al., 2019): high-pass filter ECG at 30 Hz, blank QRS (±50 ms around R-peaks). The residual reveals inspiratory EMG bursts. If no bursts detected → no effort → central [6].

11. Hypoxic Burden

11.1 Clinical Motivation

The AHI counts events but ignores their physiological impact. Two patients with the same AHI can have radically different cardiovascular risk depending on the *depth* and *duration* of their desaturations. The hypoxic burden captures this by integrating the total oxygen deficit [5].

11.2 Algorithm

For each respiratory event i:

1. Compute pre-event SpO_2 baseline: 90th percentile in a 120-second window, with global 95th percentile fallback.
2. Integrate desaturation area: $\text{area}_i = \int_{t_{\text{onset}}}^{t_{\text{recovery}}} \max(0,\ B_{\text{SpO}_2} - \text{SpO}_2(t))\, dt$
3. Normalise: $\text{HB} = \sum_i \text{area}_i \,/\, (60 \times \text{TST}_{\text{hours}})$ [%·min/h]

Table 4: Hypoxic burden risk thresholds (Azarbarzin et al. 2019)

Risk	HB (%·min/h)	Clinical
Low	<20	Low CV risk
Moderate	20–73	Moderate risk
High	>73	HR 1.47 for incident CVD

```
from psgscoring import compute_hypoxic_burden
result = compute_hypoxic_burden(spo2_data, sf_spo2=1.0,
                                resp_events=events, hypno=hypno)
print(f"HB: {result['hypoxic_burden']} {result['unit']}")
```

11.3 Validation: The Pevernagie Phenotype

On PSG-IPA, SN2 (AHI 8.9, Mild) has HB 71.7—approaching the high-risk threshold. SN1 (AHI 7.6, also Mild) has HB 12.5. Same AHI category, 6× difference in oxygen deficit. On iSLEEPS ($n = 96$), 9 patients with AHI <15 had HB >40, including one with AHI 9.0 and HB 114.

12. Post-Processing: Central Apnea Refinement

After the main detection pipeline, three refinements are applied automatically (pipeline Step 11).

12.1 CSR-Aware Central Reclassification

During Cheyne-Stokes respiration, cardiac pulsation on the effort bands mimics respiratory effort, causing central apneas to be misclassified as obstructive. Events with `csr_flagged=True` that are typed `"obstructive"` or `"mixed"` are reclassified as `"central"`.

Validation: PSG-IPA: 23 events reclassified. iSLEEPS ($n = 96$): 365 events in 36/96 studies (37%).

12.2 Mixed Apnea Decomposition

Each mixed apnea is decomposed into central (absent effort) and obstructive (resumed effort) portions. If the central portion ≥10 s, the event is reclassified as central.

```
from psgscoring import decompose_mixed_apneas
events = decompose_mixed_apneas(events, thorax, abdomen, sf)
# Each mixed event gains: central_duration_s,
#   obstructive_duration_s, central_ratio
```

12.3 Central Instability Index (CII)

Quantifies profile-dependent O/C uncertainty by comparing AHI across profiles. CII <0.15 = stable; 0.15–0.40 = moderate; >0.40 = high (manual review recommended).

13. Arousal Detection

AASM criteria: abrupt EEG frequency shift (≥3 s, preceded by ≥10 s stable sleep; during REM: concurrent EMG increase).

Phase 1 (EEG-based): sliding-window alpha+beta power exceeding mean + 2×SD. Phase 2 (filtering): K-complex exclusion (raises minimum duration from 3 s to 5 s), CVR confidence boost (≥10 bpm swing within 15 s), respiratory coupling (within 3 s of event termination → respiratory arousal), Rule 1B reinstatement (flow reduction + arousal without desaturation → hypopnea).

14. Ancillary Analyses

Snoring: bandpass 30–300 Hz, 1-second RMS windows, threshold at P60 of RMS during sleep.

Heart rate: R-peak detection via bandpass 5–30 Hz + `find_peaks()`. Metrics: mean/min/-max HR, bradycardia/tachycardia episodes.

Body position: auto-mapping of ADC/integer/string encodings. Positional AHI per position. Supine-dominant OSA: supine AHI ≥ 2× non-supine. Position-induced baseline reset at transitions.

Cheyne-Stokes: autocorrelation of flow envelope over 5-minute windows; secondary peak at 40–100 s confirms CSR periodicity.

PLM: AASM 2.6 + WASM criteria (0.5–10 s duration, ≥8 μV, 5–90 s interval, ≥4 consecutive). Respiratory-associated movements excluded from PLMI.

Sleep cycles: Feinberg & Floyd criteria (NREM ≥15 min, REM consolidation with 2-min gap tolerance). Stage-specific AHI (REM vs. NREM) identifies REM-dominant phenotype.

15. Signal Quality Assessment

Every channel is evaluated on four dimensions: flat-line percentage (rolling 2-second SD), clipping (>10 consecutive samples at ADC rail), line-noise dominance (49–51 and 59–61 Hz), and disconnect events. Each channel receives a grade (good/acceptable/poor/unusable).

Montage plausibility: EEG↔EOG correlation >0.95 suggests shared reference; thorax↔abdomen >0.98 suggests duplication.

16. The Complete Pipeline

`run_pneumo_analysis()` orchestrates 11 steps:

Table 5: Pipeline steps

Step	Description
1	Signal quality assessment
2	Linearisation + filtering + envelope
3	Dynamic baseline computation
4	Apnea detection (thermistor, ≥90%)
5	Hypopnea detection (nasal pressure, ≥30%)
6	SpO_2 coupling + type classification
7	6 over-counting corrections
8	6 under-counting corrections + stability filter
9	AHI confidence interval (3 profiles)
10	Hypoxic burden (Azarbarzin 2019)
11	Post-processing (CSR recl. + mixed decomp. + CII)

17. External Validation

17.1 PSG-IPA (5 recordings, 60 scorer sessions)

Table 6: PSG-IPA: algorithm vs. 12 scorers per recording

	n	$\mathbf{AHI_{med}}$	$\mathbf{AHI_{algo}}$	$\lvert\Delta\rvert$	**O**	**C**	**HB**	**CSR recl.**
SN1	11	6.0	7.6	1.6	9	7	12.5	7
SN2	12	3.7	8.9	5.2	3	0	**71.7**	0
SN3	12	54.0	53.5	0.5	252	30	**241.8**	16
SN4	12	3.5	4.2	0.7	0	0	22.1	0
SN5	12	9.9	10.7	0.8	17	1	37.0	0
Mean $\lvert\Delta\rvert$: 1.8/h								23 total

Event-level matching on SN3 (severe): F1= 0.890, mean onset difference 2.3 s. For 3/5 recordings, algorithm deviation was smaller than inter-scorer variability.

17.2 iSLEEPS (96 ischemic stroke patients)

Table 7: iSLEEPS by severity ($n = 96$, 1 outlier excluded)

	Overall ($n = 96$)	**Normal** ($n = 12$)	**Mild** ($n = 20$)	**Mod.** ($n = 26$)	**Severe** ($n = 38$)
Bias	−14.3/h	+1.3	−1.3	−12.4	−27.5
MAE	15.6/h	3.4	3.9	13.0	27.5
r	0.653				
Conc.	30%	83%	45%	23%	11%

365 CSR reclassifications in 36/96 studies. 15 patients with HB >73 (high CV risk). 9 patients with AHI <15 but HB >40.

18. Data Structures

18.1 Event Dictionary

```
{
    "type": "obstructive",  # or central/mixed/hypopnea
    "onset_s": 1234.5,
    "duration_s": 18.0,
    "stage": "N2",
    "desaturation_pct": 4.2,
    "min_spo2": 89.3,
    "confidence": 0.82,
    "rule_index": 1,
    "flow_reduction": 0.45,
    "classify_detail": { ... },
    "csr_flagged": False,
    "spo2_cross_contaminated": False,
    # Post-processing (v0.2.93):
    "csr_reclassified": True,
    "original_type": "obstructive",
    "central_duration_s": 12.1,
    "obstructive_duration_s": 7.9,
    "central_ratio": 0.60,
    "mixed_decomposed": True,
}
```

18.2 Hypoxic Burden Output

```
results["hypoxic_burden"] = {
    "hypoxic_burden": 71.7,        # %*min/h
    "n_events_with_burden": 40,
    "mean_event_burden": 645.3,    # %*s per event
    "unit": "%*min/h",
}
```

18.3 Post-Processing Output

```
results["postprocess"] = {
    "n_csr_reclassified": 7,
    "n_mixed_decomposed": 3,
    "n_mixed_to_central": 2,
```

```
    "central_instability": {
        "central_instability_index": 0.43,
        "interpretation": "high",
        "ahi_range": 3.3,
    },
    "cai_change": +9,
}
```

19. Practical Tutorial

```python
import mne
from psgscoring import run_pneumo_analysis

raw = mne.io.read_raw_edf("patient.edf", preload=True)
hypno = ["W","N1","N2","N3","R", ...]  # per 30-s epoch

results = run_pneumo_analysis(raw, hypno, channel_map={
    "flow_pressure": "Pressure",
    "flow_thermistor": "Flow",
    "thorax": "RIP Thora", "abdomen": "RIP Abdom",
    "spo2": "SpO2", "ecg": "ECG II",
}, scoring_profile="standard")

# AHI with confidence interval
iv = results["ahi_interval"]
print(f"AHI: {iv['standard']['ahi']}")
print(f"[{iv['strict']['ahi']}"
      f"-{iv['sensitive']['ahi']}] "
      f"Grade {iv['robustness_grade']}")

# Hypoxic burden
hb = results["hypoxic_burden"]
print(f"HB: {hb['hypoxic_burden']} {hb['unit']}")

# Post-processing
pp = results["postprocess"]
print(f"CSR reclassified: {pp['n_csr_reclassified']}")
```

20. Design Decisions

Why rules, not deep learning? Transparency (every event traceable to AASM rule), data efficiency (zero training data), auditability (CE/FDA), correction specificity (each mechanism independently addressable). The planned hybrid uses LightGBM *on top of* the rule-based engine.

Performance: 3–5 min for an 8-hour PSG at 256 Hz. All 12 corrections add <1 s overhead.

21. Common Pitfalls

Variable scoping: pass computed variables explicitly to inner functions—Python closures capture by reference. **Docker caching:** `docker compose restart` does not rebuild; use `build -no-cache`. **Shell escaping:** use Python parameterised queries for passwords (bcrypt $ interpretation). **EDF:** pyedflib for EDF+ writing; edfio silently drops annotations. **Translation shadowing:** never `t = date.today()` when `t()` is imported.

22. Exercises

1. Generate a synthetic nasal pressure signal (12 breaths/min). Apply square-root linearisation. Compare the envelope before and after.
2. Create a 10-minute signal with a 30-second apnea and 150% recovery breathing. Plot the dynamic baseline with and without the 30-second recovery mask (Fix 1).
3. Create test inputs for `classify_apnea_type()` that trigger each of the 7 rules.
4. Generate a SpO_2 signal with a 5% desaturation at $t = 40$ s. Test `get_desaturation()` with 30 s and 45 s windows.
5. Download a public PSG from PhysioNet. Run all three scoring profiles and compare AHI.
6. Generate a SpO_2 signal with three desaturations (3%, 8%, 15%). Call `compute_hypoxic_burden()` and verify the deep event dominates.
7. Create 10 events with `csr_flagged=True` and type `"obstructive"`. Call `reclassify_csr_events()` and verify reclassification.
8. Generate a 20-second effort signal: 12 s silence then 8 s sinusoidal. Call `decompose_mixed_apneas()` and verify central reclassification.
9. Using the iSLEEPS CSV, identify patients where AHI <15 but HB >40. How does hypoxic burden change the clinical picture?

Glossary

AHI	Apnea-Hypopnea Index
OAHI	Obstructive AHI
CAI	Central Apnea Index
RDI	Respiratory Disturbance Index (AHI + RERA index)
RERA	Respiratory Effort-Related Arousal
ODI	Oxygen Desaturation Index
HB	Hypoxic Burden (%·min/h)
CII	Central Instability Index
PLMI	Periodic Limb Movement Index
TST	Total Sleep Time
RIP	Respiratory Inductance Plethysmography
TECG	Transformed ECG (high-pass + QRS-blanked)
CSR	Cheyne-Stokes Respiration
MMSD	Mean Magnitude of Second Derivative
EDF	European Data Format
AASM	American Academy of Sleep Medicine

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```
pip install psgscoring
https://github.com/bartromb/psgscoring
https://slaapkliniek.be
```